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Amorphous-rich $RuMnO_x$ aerogel with weakened Ru–O covalency for efficient acidic water oxidation

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ABSTRACT

Ruthenium dioxide (RuO_2) is one of the most promising acidic oxygen evolution reaction (OER) catalysts to replace the expensive and prevalent iridium (Ir)-based materials. However, the lattice oxygen oxidation induced Ru dissolution during OER compromises the activity and stability. Amorphous materials have been identified as a viable strategy to promote the stability of RuO_2 in acidic OER applications. This study reported a nanoporous amorphous-rich $RuMnO_x$ (A-$RuMnO_x$) aerogel for efficient and stable acidic OER. Compared with highly crystalline $RuMnO_x$, the weakened Ru–O covalency of A-$RuMnO_x$ by forming amorphous structure is favorable to inhibiting the oxidation of lattice oxygen. Meanwhile, this also optimizes the electronic structure of Ru sites from overoxidation and reduces the reaction energy barrier of the rate-determining step. As a result, A-$RuMnO_x$ aerogel exhibits an ultra-low overpotential of 145 mV at 10 mA cm^{-2} and durability exceeding 100 h, as well as high mass activity up to 153 mA mg_{Ru}^{-1} at 1.5 V vs. reversible hydrogen electrode (RHE). This work provides valuable guidance for preparing highly active and stable Ru-based catalysts for acidic OER.

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1. Introduction

The electrolysis of water combined with renewable energy is one of the most promising ways to produce hydrogen [1–3]. However, the oxygen evolution reaction (OER) involves multiple electron transfer processes, resulting in significant energy consumption, which severely limits practical applications [4,5]. Among the numerous water electrolysis technologies, proton exchange membrane (PEM) water electrolysis has attracted extensive research interest due to low impedance, high current density, and fast response speed, but the activity and sustainability of acidic OER electrocatalysts are still unsatisfactory and need further improvement [6,7]. Notably, ruthenium dioxide (RuO_2), which has the advantages of low cost and adjustable electronic structure, is expected to be an ideal alternative to iridium (Ir)-free acidic OER catalysts [8–11]. Unfortunately, the activity and stability of RuO_2 under harsh acidic conditions have become major obstacles [12,13]. In view of this, the introduction of heteroatoms into RuO_2 is a simple and efficient way to regulate the electronic structure of Ru, thereby affecting activity and stability [14–19]. Transition metal Mn, which has good acid resistance and can affect the binding with oxo-intermediates, has become a common doping element [9,20–23]. Although RuMn-based catalysts have been extensively studied, their activity and stability remain limited. Consequently, the design of RuMn-based OER electrocatalysts to overcome the "see-saw" relationship between activity and stability under acidic conditions remains a huge challenge worthy of further exploration.

Recent studies have shown that high Ru–O covalency often triggers lattice oxygen mechanisms (LOM) during the OER, leading to the formation of oxygen vacancies (O_v) to dissolve Ru (soluble RuO_4 by overoxidation), which ultimately accelerates the deactivation of the catalyst [24–28]. Despite considerable efforts to stabilize Ru-based electrocatalysts, there is still a significant

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performance gap in meeting industrial requirements. Thus far, two main strategies have been proposed to improve the stability of Ru-based electrocatalysts. The lattice oxygen is stabilized by weakening the covalency of Ru–O to reduce the number of O_v [10,29,30]. For example, Zhang et al. introduced W and Er into the lattice of RuO_2 to improve the stability of acidic OER [26]. Guo et al. prepared lanthanide-dope RuO_2 to modulate the Ru–O bond covalency for enhancing stability [22]. However, very few elements have the capability to directly weaken the covalency of the Ru–O bond to improve stability. Therefore, there is an urgent need to explore effective strategies for weakening the covalency of Ru–O bond. Unlike crystalline materials, amorphous materials have unique coordination and flexible structures. It has been reported that amorphous metal oxides usually exhibit lattice expansion and give lower density than crystals [31–33]. The lattice expansion with lengthened M–O bonds can help to weaken the M–O covalency [34].

On the other hand, enhancing the intrinsic OER activity by lowering the required overpotential is also helpful for mitigating the over-oxidation of Ru species during OER [35,36]. To this end, extensive efforts have been made to increase active sites and regulate intermediate species, including strain engineering, heterophase structures, and morphology design [30,37,38]. Among them, porous structures have received extensive attention in electrocatalysis due to small diffusion resistance and short diffusion paths [39–41]. The high porosity and large specific surface area provide an abundance of catalytically active centers [42]. Therefore, the clever construction of amorphous structures in porous RuMn-based materials not only provides a large number of active centers to enhance the intrinsic OER activity, but also weakens the covalency of Ru–O, which may provide an opportunity to achieve excellent electrocatalytic properties.

Herein, this study reports the amorphous-rich $RuMnO_x$ (denoted as A-$RuMnO_x$) aerogel electrocatalyst obtained from the oxidation of porous RuMn aerogels to boost the OER activity and stability. The porous structure of A-$RuMnO_x$ provides accessible active centers to promote the OER activity. In addition, the presence of amorphous phase weakens the covalency of Ru–O to alleviate the involvement of LOM, improving the activity and stability. Among all of the samples, A-$RuMnO_x$ exhibits an excellent acidic OER performance, requiring only 145 mV overpotential at 10 mA cm^{-2}, which is considerably lower than the as-prepared RuO_2 (221 mV), commercial RuO_2 (C-RuO_2, 271 mV), commercial IrO_2 (C-IrO_2, 487 mV), nanoparticles $RuMnO_x$ (NP-$RuMnO_x$, 225 mV), and highly crystalline $RuMnO_x$ ($RuMnO_x$, 182 mV). Furthermore, the introduction of Mn not only reduces the Ru content but also greatly increases the intrinsic catalytic activity, making A-$RuMnO_x$ achieve a significantly high mass activity of 153 mA mg_{Ru}^{-1} at 1.5 V vs. reversible hydrogen electrode (RHE). In-situ X-ray diffraction (XRD) and in-situ Raman spectroscopy confirmed the high acidic OER stability of A-$RuMnO_x$. Similarly, in-situ attenuated total reflection-surface enhanced infrared absorption spectroscopy (ATR-SEIRAS) revealed that A-$RuMnO_x$ allows the adsorbate evolution mechanism (AEM), which in combination with density functional theory (DFT) calculations confirmed the optimized adsorption of the oxygenated intermediates in the OER. This work shows that the structure design of the catalyst helps to balance the relationship between activity and stability, which provides ideas for the development of more stable and efficient acidic electrocatalysts.

2. Experimental

2.1. Materials

Manganese chloride tetrahydrate ($MnCl_2{\cdot}4H_2O$), ruthenium chloride Hydr ($RuCl_3{\cdot}H_2O$), and sodium borohydride ($NaBH_4$) were purchased from Aladdin (Shanghai, China). Commercial ruthenium oxide (C-RuO_2) and iridium oxide (C-IrO_2) were purchased from InnoChem (Beijing, China). Hydrazine hydrate (N_2H_4 H_2O) was purchased from Sinopharm. Nafion® D-520 was provided by Alfa Aesar (Beijing, China). All chemical reagents used were not further purified.

2.2. Sample preparation

2.2.1. Synthesis of RuO_2

$RuCl_3{\cdot}H_2O$ (0.0105 mg) was uniformly dispersed into H_2O (190 mL). Subsequently, freshly prepared $NaBH_4$ aqueous solution (1 mol L^{-1}, 5.8 mL) was rapidly injected. After standing for 5 h, the Ru aerogel was obtained by centrifugal washing and freezing drying. RuO_2 aerogel can be prepared by calcining Ru aerogel at 400 °C for 0.5 h in muffle furnace.

2.2.2. Synthesis of A-$RuMnO_x$

$RuCl_3{\cdot}H_2O$ (0.0105 mg) and $MnCl_2{\cdot}4H_2O$ (0.0023 mg) were uniformly dispersed into H_2O (190 mL). Subsequently, freshly prepared $NaBH_4$ aqueous solution (1 mol L^{-1}, 5.8 mL) was rapidly injected. After standing for 5 h, the RuMn aerogel was obtained by centrifugal washing and freezing drying. A-$RuMnO_x$ aerogel can be prepared by calcining RuMn aerogel at 400 °C for 0.5 h in muffle furnace.

2.2.3. Synthesis of $RuMnO_x$ with different Ru and Mn contents

Similar to the preparation of A-$RuMnO_x$, only $MnCl_2{\cdot}4H_2O$ becomes 0.0012 mg or 0.0046 mg, labeled as $RuMnO_x$-1 and $RuMnO_x$-2.

2.2.4. Synthesis of $RuMnO_x$-300 °C and synthesis of $RuMnO_x$-500 °C

Similar to the preparation method of A-$RuMnO_x$, only the calcination temperature was changed (300 and 500 °C).

2.2.5. Synthesis of NP-$RuMnO_x$

Similar to the synthesis of A-$RuMnO_x$, only $NaBH_4$ was replaced with $N_2H_4{\cdot}H_2O$.

2.2.6. Synthesis of $RuMnO_x$

Similar to the synthesis of A-$RuMnO_x$, the heat treatment time was only increased to 10 h.

2.3. Instrument

The crystal structure of the catalyst was tested by XRD with Cu-K_α radiation (λ = 1.54 Å). The morphology was characterized by transmission electron microscope (TEM) with 200 kV field emission gun (Japan), scanning electron microscope (SEM) (Hitachi, SU8010, Tokyo, Japan) at 15 kV, and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM). X-ray photoelectron spectroscopy (XPS) was recorded using a WSCAL-ab 220i-XL spectrometer and a monochromatic Al-K_α radiation source. The element distribution and content of the catalyst were tested using the energy-dispersive X-ray spectrometry (EDS) with an IXRF SDD 2610 EDS system and inductively coupled plasma (ICP) with an avio™ 200 ICP-OES. Raman spectra were obtained on an InVia 1WU072 Raman spectrometer with λ = 532 nm. Fourier transform infrared (FT-IR) spectrometer was obtained on a Bruker INVENIO R. Brunauer-Emmett-Teller (BET) surface areas were measured with a Micromeritics TriStar II 3020 instrument based on N_2 adsorption-desorption isotherms recorded at 77 K. Electron paramagnetic resonance (EPR) spectra were collected on a Bruker EMXPLUS10.

2.4. Electrochemical measurements

All the electrochemical tests were implemented on the Princeton electrochemical workstation (PARSTAT MC, Princeton, USA) with a standard three-electrode system for OER in 0.5 M H_2SO_4. The carbon paper (CP) (1 cm × 1 cm) or glassy carbon (GC), Hg/HgSO$_4$, and carbon rod acted as the working electrode, reference electrode, and auxiliary electrode, respectively. Linear sweep voltammetry (LSV) polarization curves were conducted with 2 mV s^{-1} of scan rate. Electrochemical impedance spectroscopy (EIS) curves were obtained in the frequency range from 100 kHz to 0.05 Hz at 1.45 V. The electrochemically active surface areas (ECSA) were acquired from the double-layer capacitance (C_{dl}) according to the formula: ECSA = C_{dl}/C_s, where C_s is the specific capacitance of the sample, generally C_s = 0.06 mF cm^{-2} for oxide surface in H_2SO_4. Reversible hydrogen electrode (RHE) and Tafel slope were calculated according to the following formula: $E_{RHE} = E_{Hg/Hg2SO4} + 0.652 + 0.059$ pH (V); $\eta = a + b \times \log j$, where η, b, and j represent overpotential, Tafel slope, and current density, respectively. All electrochemical data were *iR* compensated and converted to RHE.

3. Results and discussion

The amorphous-rich A-RuMnO$_x$ aerogel was prepared by a simple reduction-oxidarion two-step method (Fig. S1). Comprehensive characterization techniques were used to detect changes in crystal structure throughout the preparation process. XRD shows that the RuMn aerogel binary alloy exhibits an amorphous structure (Fig. S2). After the heat treatment, XRD results reveal that the diffraction peaks of the synthesized RuO_2, A-RuMnO$_x$, NP-RuMnO$_x$, and RuMnO$_x$ could be indexed to rutile structured RuO_2 (JCPDS No. 43-1027), without obvious diffraction peaks related to MnO_x or without significant peak shifts, suggesting the successful incorporation of Mn atoms into RuO_2 (Fig. 1a, Figs. S3 and S4) [22]. Notably, the peak intensity of the A-RuMnO$_x$ increases upon extending the annealing time, which indicates that the crystallinity increased. In addition, the peak corresponding to A-RuMnO$_x$ shifts to a lower diffraction angle, confirming the generation of lattice stretching (Fig. S5) [41,43,44]. Furthermore, Raman results confirm that these catalysts have similar spectra, while the Raman peaks (E_g, A_{1g}, and B_{2g}) shift towards lower wavenumbers from RuMnO$_x$ to A-RuMnO$_x$, which indicates that the vibrational frequency of Ru–O decreased and the Ru–O bond length increased in A-RuMnO$_x$ (Fig. 1b, Figs. S6 and S7) [45,46].

The chemical states and elemental composition of the prepared catalyst were studied by XPS. The full XPS spectrum confirms the presence of Mn (Fig. S8). Fig. 1(c) and Fig. S9 and S10 show the XPS Ru 3*d* and C 1*s* spectra, where the peaks at 284.80 eV correspond to C=C (deconvoluted into C=C and C–O bands) and Ru 3$d_{3/2}$ due to the partial overlap of the peak signals of C 1*s* and Ru 3*d* [44]. For the Ru 3$d_{5/2}$ spectra in RuMnO$_x$, this can be deconvoluted into two peaks at 281.02 eV (Ru^{4+}) and 282.23 eV (satellite peak) [47]. Obviously, Ru^{4+} (280.71 eV) and satellite peak (281.91 eV) for A-RuMnO$_x$ move towards lower binding energy relative to RuMnO$_x$, implying a slightly lower Ru oxidation state in A-RuMnO$_x$ (Fig. 1c). However, Ru 3$d_{5/2}$ in NP-RuMnO$_x$ is located at the highest binding energy, indicating that Ru has a relatively high oxidation state, which impairs the acidic stability (Fig. S9). Moreover, considering the higher electronegativity of Mn ions compared to Ru ions, there is a slight positive shift in the bond energy of Ru 3$d_{5/2}$ in RuMnO$_x$, which activates the Ru sites (Fig. S10) [21]. In the Mn 2*p* spectrum of RuMnO$_x$, Mn 2$p_{3/2}$ and Mn 2$p_{1/2}$ were resolved to Mn^{2+} (641.43/652.98 eV) and Mn^{3+} (643.13/654.50 eV), respectively [48,49]. By comparison, the peaks of Mn^{2+} (640.90/652.60 eV) and Mn^{3+} (642.46/654.03 eV) in A-RuMnO$_x$ catalyst also tended to lower binding energy, indicating that Mn has a relatively low oxidation state (Fig. 1d). The electronic properties and valence were further studied by X-ray absorption spectroscopy (XAS). The Ru *K*-edge X-ray absorption near-edge structure (XANES) spectra indicate that Ru in A-RuMnO$_x$ is similar to standard RuO_2, verifying that the valence state of Ru is about +4 (Fig. 1e). However, the adsorption edge of A-RuMnO$_x$ is slightly lower than that of standard RuO_2, indicating a relatively lower Ru oxidation state (inserted image of Fig. 1e). Furthermore, the Mn *K*-edge XANES spectra show that the adsorption edge of A-RuMnO$_x$ is close to that of standard Mn_3O_4. These valence results are consistent with the above XPS results (Fig. 1f). Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) measurement was performed to further understand the local chemical environment of the catalyst. As shown in Fig. 1(g), the Ru–O bond length of A-RuMnO$_x$ is longer than that of RuO_2. The lower Ru valence state and elongated Ru–O bond further demonstrated that the Ru–O bond covalency was weakened by introducing the amorphous structure [35,50]. Simultaneously, no obvious Ru–Ru and Mn–Mn bonds were observed in A-RuMnO$_x$, indicating that most of the metals in the catalyst were oxidized (Fig. 1h).

The specific surface area and pore distribution of the prepared catalysts were analyzed using N_2 adsorption/desorption isotherms. The BET results verify that A-RuMnO$_x$ (108.40 m^2 g^{-1}) inherits the inherent high specific surface area and mesoporous structure (2–50 nm) of the RuMn aerogel, which is higher than both RuMnO$_x$ (89.18 m^2 g^{-1}) and NP-RuMnO$_x$ (28.64 m^2 g^{-1}) (Fig. 1i, Figs. S11–S13). The significant increase in the specific surface area provides abundant adsorption sites, which is conducive to improving the catalytic performance of the catalyst [51].

The morphology and structure of the catalyst were verified by SEM and TEM. SEM confirms that the morphology of RuO_2, A-RuMnO$_x$, and RuMnO$_x$ is similar to that of RuMn aerogel, while NP-RuMnO$_x$ exhibits a granular morphology (Fig. S14). TEM further reveals that the morphology is comprised of a large number of interconnected nanoparticles with sizes of about 10 nm. For comparison, NP-RuMnO$_x$ has an obvious agglomeration spatial structure (Fig. 2a, g and Fig. S15). High-resolution TEM (HRTEM) and selective area electron diffraction (SAED) patterns show that the lattice spacing of 0.319, 0.256, and 0.166 nm could be attributed to the (110), (101), and (211) crystal planes of A-RuMnO$_x$ (Fig. 2b–d), which are slightly larger than that of homemade RuO_2 (Fig. S16), RuMnO$_x$ (Fig. 2h and i), and NP-RuMnO$_x$ (Fig. S17). This is consistent with the XRD, Raman, and XAS results, providing more direct evidence for the lattice strain in A-RuMnO$_x$. Furthermore, HRTEM also reveals the presence of amorphous phases in A-RuMnO$_x$, which occupy the space between the crystals (Fig. 2b). HAADF-STEM distinguish the atomic arrangement in the different regions and the regular arrangement of atoms in most of the (110) and (101) crystal regions, indicating the presence of a crystalline structure. Conversely, the atoms in the other regions are randomly arranged, proving the presence of an amorphous structure (Fig. 2e and Fig. S18). HRTEM confirms that the amorphous region in RuMnO$_x$ almost disappears upon prolonging the heat treatment time, forming a crystalline phase, which indicates that the catalyst requires more energy input during the crystallization process (Fig. 2h and i). In addition, EDS mapping confirms that Ru, Mn, and O are evenly distributed, which is consistent with the addition results (Fig. 2f, Tables S1 and S2). The HRTEM and HAADF-STEM analyses show that A-RuMnO$_x$ exhibits abundant amorphous structure. The presence of more amorphous regions in A-RuMnO$_x$ induces significant stretching of Ru–O bonds, which weakens Ru–O covalency. It is widely believed that weak metal–oxygen covalency can inhibit the LOM pathway during the OER, thereby improving catalyst stability.

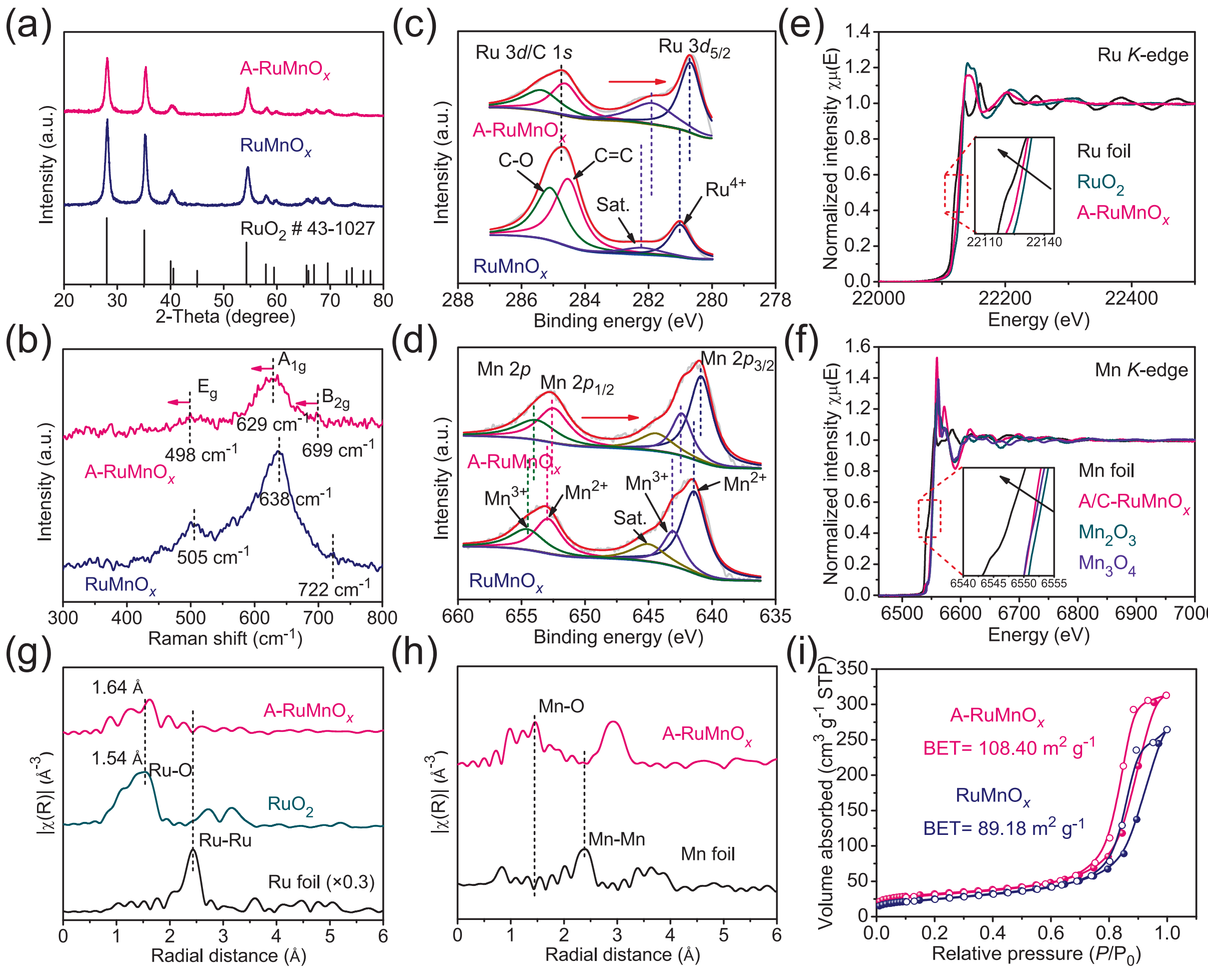

Fig. 1. (a) XRD patterns; (b) Raman spectrum; (c) Ru 3d/C 1s XPS spectrum; (d) Mn 2p XPS spectrum; (e, f) XANES spectra; (g, h) FT-EXAFS spectra; (i) nitrogen adsorption/desorption isotherms.

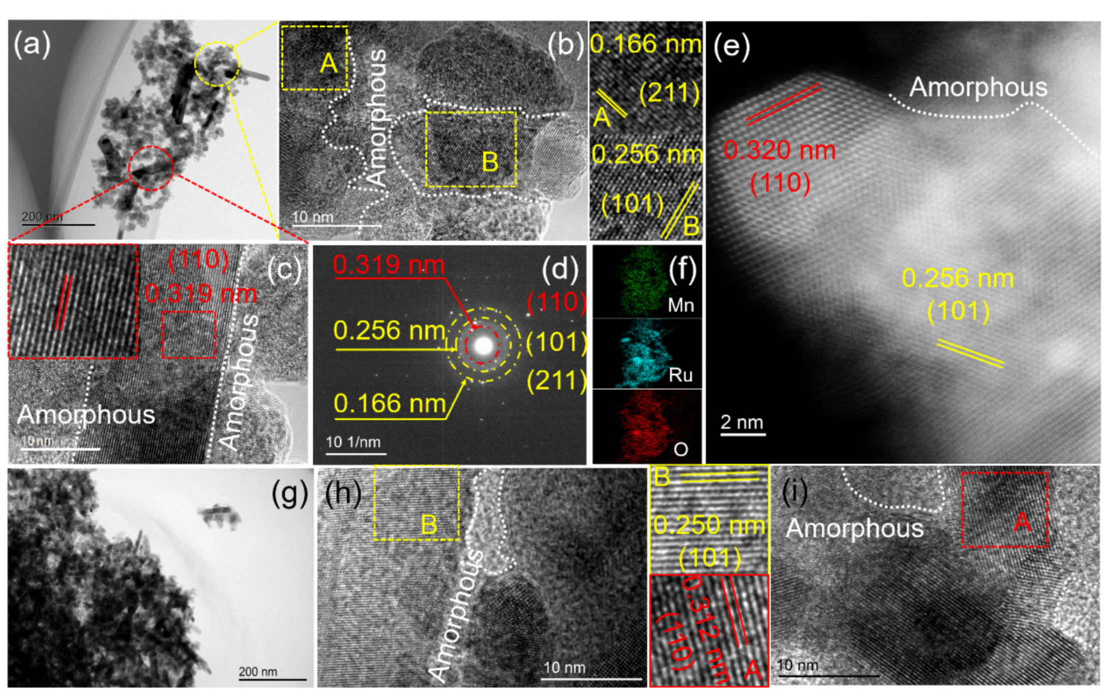

Fig. 2. (a) TEM image, (b, c) HRTEM images, (d) SAED pattern, (e) HAADF-STEM, and (f) EDS mapping of A-RuMnO$_x$; (g) TEM image and (h, i) HRTEM images of RuMnO$_x$.

To investigate the effect of these structural changes on the OER performance, the electrochemical OER performance of the A-$RuMnO_x$ aerogel was tested in 0.5 M H_2SO_4. For comparison, as-prepared RuO_2, $RuMnO_x$, RuMn aerogel, C-IrO_2, and C-RuO_2 were also evaluated. It is worth noting that the pristine RuMn aerogel undergoes a rapid deactivation process, which is unsuitable for using as an acidic OER electrocatalyst (Fig. S19). As shown in Fig. 3(a), A-$RuMnO_x$ exhibits an extremely low overpotential of 145 mV at 10 mA cm^{-2}, which is superior to RuO_2 (221 mV) synthesized by the same method, $RuMnO_x$ (182 mV), C-RuO_2 (271 mV), and C-IrO_2 (487 mV), indicating that excellent catalytic activity can be achieved by structural design. Moreover, the overpotentials at 10 mA cm^{-2} are compared with previous noble metal catalysts, which confirm that the A-$RuMnO_x$ aerogel maintains high activity (Fig. 3g and Table S3). We also performed O_2-saturation experiments to confirm the OER activity of A-$RuMnO_x$ (Fig. S20). Meanwhile, a control sample comprised of NP-$RuMnO_x$ without pores is prepared via a hydrazine hydrate reduction in order to elucidate the promotional role of the porous structure. NP-$RuMnO_x$ (225 mV) requires a larger overpotential than A-$RuMnO_x$, which highlights the important role of the porous structure in promoting OER activity (Fig. S21). This kinetic behavior was supported by the Tafel slope values, as shown in Fig. 3(b) and Fig. S22. A-$RuMnO_x$ (62.7 mV dec^{-1}) exhibits a lower Tafel slope than RuO_2 (90.1 mV dec^{-1}), $RuMnO_x$ (68.6 mV dec^{-1}), and NP-$RuMnO_x$ (89.5 mV dec^{-1}), which could be possibly attributed to the optimal crystal structure, improving the reaction kinetics. Moreover, A-$RuMnO_x$ still has low overpotential and Tafel slope at high current density, confirming excellent activity and kinetic processes (Fig. S23). Subsequently, EIS was also used to verify the charge transfer kinetics of OER. The charge transfer impedance (R_{ct}) of A-$RuMnO_x$ (2.56 Ω) is lower than that of self-made RuO_2, $RuMnO_x$, NP-$RuMnO_x$, as well as commercial RuO_2 and IrO_2 (Fig. 3c, Fig. S24 and Table S4).

Previous studies have shown that enhanced activity can be attributed to an increase in intrinsic activity and the number of active sites [52]. The OER activity was further elucidated by measuring the double-layer capacitance (C_{dl}) and electrochemical surface area (ECSA) (Fig. S25). It can be seen that the A-$RuMnO_x$ has high C_{dl} and ECSA values, slightly higher than $RuMnO_x$, which may be attributed to the unique crystal structure with more active centers exposed (Fig. 3d, Figs. S26 and S27). The LSV curves concerning the ECSA are shown in Fig. 3(e) and Fig. S28. The normalized LSV curves show that A-$RuMnO_x$ has a higher specific activity than the control sample. A-$RuMnO_x$ also exhibits excellent mass activity, reaching 153 mA mg_{Ru}^{-1} at 1.5 V vs. RHE, which is significantly better than that of RuO_2 (35 mA mg_{Ru}^{-1}) and $RuMnO_x$ (105 mA mg_{Ru}^{-1}) (Fig. 3f). In addition, A-$RuMnO_x$ yields a turnover frequency (TOF) of 0.040 s^{-1}, which is higher than $RuMnO_x$ (0.027 s^{-1}) and RuO_2 (0.009 s^{-1}) (Table S5). The above results indicate that the structural design contributes to enhancing intrinsic activity. The effect of the heat treatment temperature and metal content on the catalyst performance was also studied. Both lower (300 °C) and

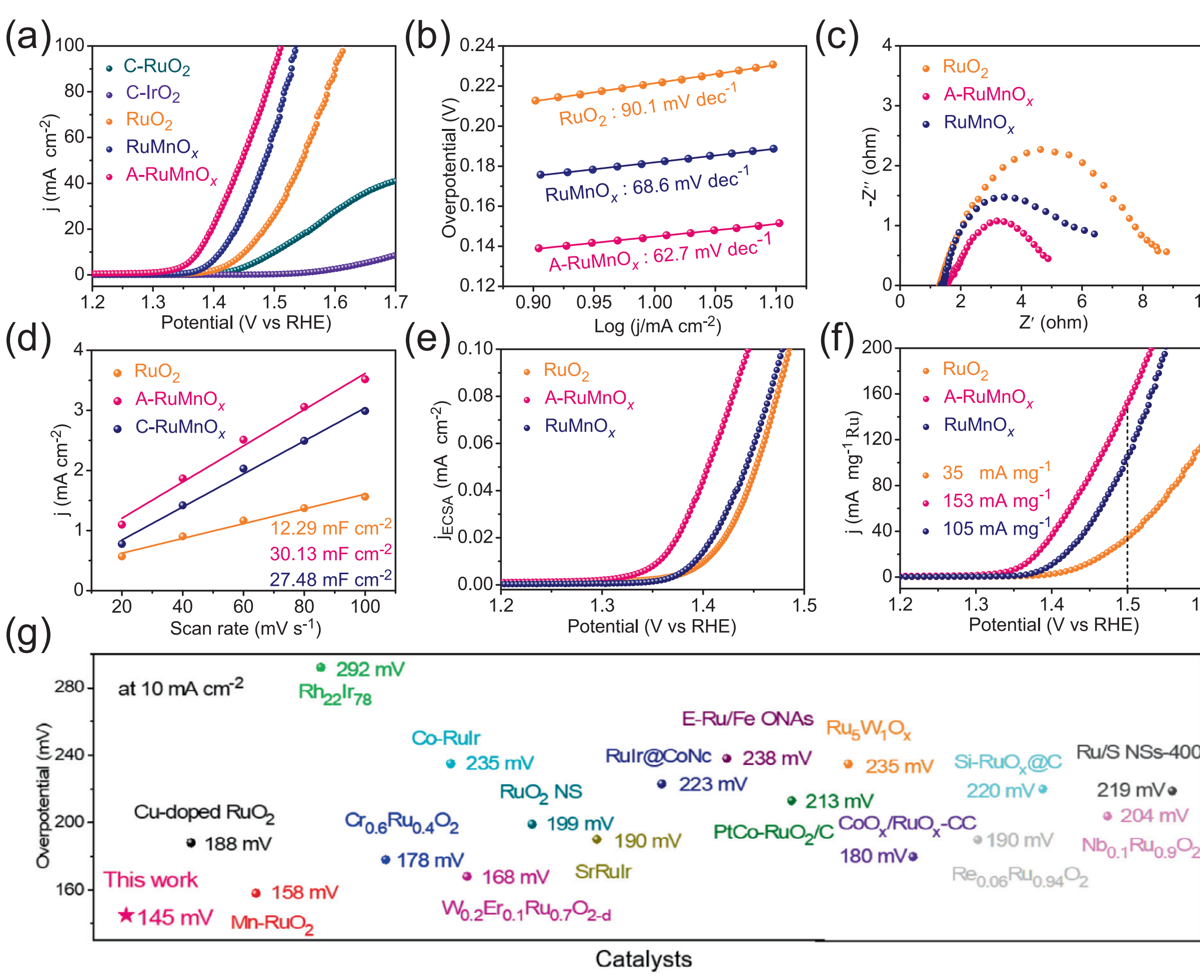

Fig. 3. (a) LSV curves, (b) Tafel plots, (c) EIS plots, (d) C_{dl} values, (e) specific activity, and (f) mass activity of homemade RuO_2, A-$RuMnO_x$, and $RuMnO_x$; (g) comparison of overpotential between A-$RuMnO_x$ and the reported Ru-based catalysts at 10 mA cm^{-2}.

higher (500 °C) temperatures can lead to poor OER performance and slow kinetics, indicating that $RuMnO_x$ is highly temperature sensitive (Fig. S29). The metal content in the catalyst also significantly affects the activity and kinetics of OER (Fig. S30).

Although high OER activity has been achieved, evaluating the long-term stability of the catalyst is still critical to validate the practical application. The stability of the catalyst was evaluated by chronopotentiometry. As shown in Fig. 4(a), A-$RuMnO_x$ can be maintained at 10 mA cm^{-2} for at least 100 h, whereas C-RuO_2 and $RuMnO_x$ undergo a rapid deactivation process. A-$RuMnO_x$ still has a long durability at higher current density of 100 mA cm^{-2} (Fig. S31), which demonstrates the excellent stability of A-$RuMnO_x$. In addition, comprehensive characterization techniques were applied to investigate the structural changes of catalysts during the OER. The morphology of the A-$RuMnO_x$ aerogel remains unchanged after the OER, and Ru, Mn, and O are still uniformly distributed throughout the sample (Fig. S32). To further verify the stability of the A-$RuMnO_x$, in situ XRD was used to monitor the structure changes during the OER (Fig. S33). The diffraction peak intensity of A-$RuMnO_x$ does not change significantly when the potential is increased to 1.7 V vs. RHE, which proves that it has excellent structural stability at higher potentials (Fig. 4b). On the contrary, the XRD peaks of RuO_2 and $RuMnO_x$ decrease at higher potentials, indicating structural damage (Fig. S34). The above results indicate that the A-$RuMnO_x$ catalyst with weak Ru–O covalency improves the stability of acidic OER.

Subsequently, the surface structure dynamics of the catalyst under constant potentials was monitored by in situ Raman spectroscopy (Fig. S35). As shown in Fig. 4(c, d) and Fig. S36, SO_4^{2-} has an obvious characteristic peak near 1000 and 1100 cm^{-1} [53], and the vibration peaks of E_g and A_{1g} can also be observed in A-$RuMnO_x$, $RuMnO_x$, and NP-$RuMnO_x$. Significantly, the vibrational peaks of A-$RuMnO_x$ remain constant as the potential increases, while the peaks of the other catalysts gradually decrease, which indicates that the structural design can effectively stabilize Ru to improve the OER stability of $RuMnO_x$ [43]. Besides, the new peaks at 430 and 590 cm^{-1} could be attributed to Ru^{4+}–O bonds and Ru^{3+}–O bonds in $RuO_2{\cdot}xH_2O$, respectively [53]. In situ ATR-SEIRAS was also used to examine the catalyst surface structure and reaction intermediates in order to gain insight into the relationship between the catalyst structure and performance (Fig. S37). There is a potential-dependent peak observed between 1200 and 1250 cm^{-1}, which is related to the stretching vibrations of the *OOH species, further confirming that the catalyst followed the AEM pathway, and is also responsible for the stability of A-$RuMnO_x$ in an acidic electrolyte (Fig. 4e, f and Fig. S38) [54]. The catalytic mechanism was further examined by pH-dependent OER activity [55,56]. As shown in Fig. S39, A-$RuMnO_x$ exhibits relatively high pH-independent OER activity, while $RuMnO_x$ shows strong pH-dependent activity. The weak sensitivity to pH changes is a feature of the AEM, indicating that A-$RuMnO_x$ follows the AEM pathway during OER. In addition, there is an additional signal peak at 1000–1050 cm^{-1} of $RuMnO_x$, which corresponds to the generation of *OO intermediates involved in the LOM pathway [20]. In particular, the peak corresponding to the *OOH species appears at 1.2 V vs. RHE and undergoes a significant negative shift in A-$RuMnO_x$. This indicates that *OOH adsorption is weakened, and thus the adsorption path is optimized and the energy barrier of the rate-determining step (RDS) from *O to *OOH is reduced, which is consistent with the DFT calculation (Fig. 5d) [8]. In order to better understand the reaction mechanism, Fig. S40 presents different reaction pathways. The A-$RuMnO_x$ follows the AEM pathway in the OER, accompanied by a multi-step adsorption process and a lowering of the adsorption energy barrier in step 3 (*OOH adsorption), thus accelerating the reaction rate. On the contrary, $RuMnO_x$ involves LOM in OER, which leads to lattice oxygen participating in the reaction, thereby reducing structural stability. In addition, the chemical environment of the catalyst surface was explored using O 1*s* XPS. The A-$RuMnO_x$ surface contains a higher percentage of surface oxygen after OER, meaning that the A-$RuMnO_x$ surface is able to maintain more reactive oxygen species, which is essential for the enhanced OER activity (Fig. S41) [57,58].

DFT calculations further revealed the relationship between the catalytic mechanism and OER performance. The constructed model is shown in Fig. 5(a and b), and the energy profile of the OER studied is based on the AEM (Fig. 5c, Figs. S42 and S43), and the results are presented in Fig. 5(d). Significantly, the RDS for both A-$RuMnO_x$ and $RuMnO_x$ is the conversion of *O to form *OOH. The RDS energy barriers for A-$RuMnO_x$ and $RuMnO_x$ are 2.11 and 2.20 eV, respectively, indicating that the amorphous structure effectively reduces the kinetic energy barrier to improve the OER activity, which is consistent with the experimental results. Combined with the

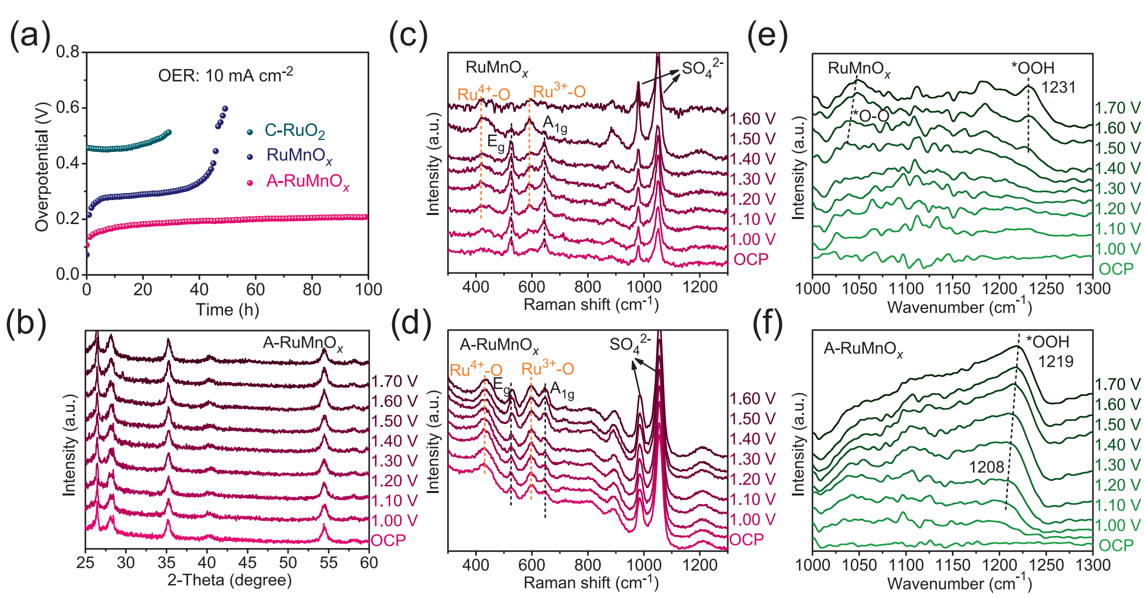

Fig. 4. (a) Stability tests of C-RuO_2, $RuMnO_x$, and A-$RuMnO_x$; (b) in-situ XRD, (c, d) in-situ Raman spectrum, and (e, f) in-situ ATR-SEIRAS of A-$RuMnO_x$ and $RuMnO_x$.

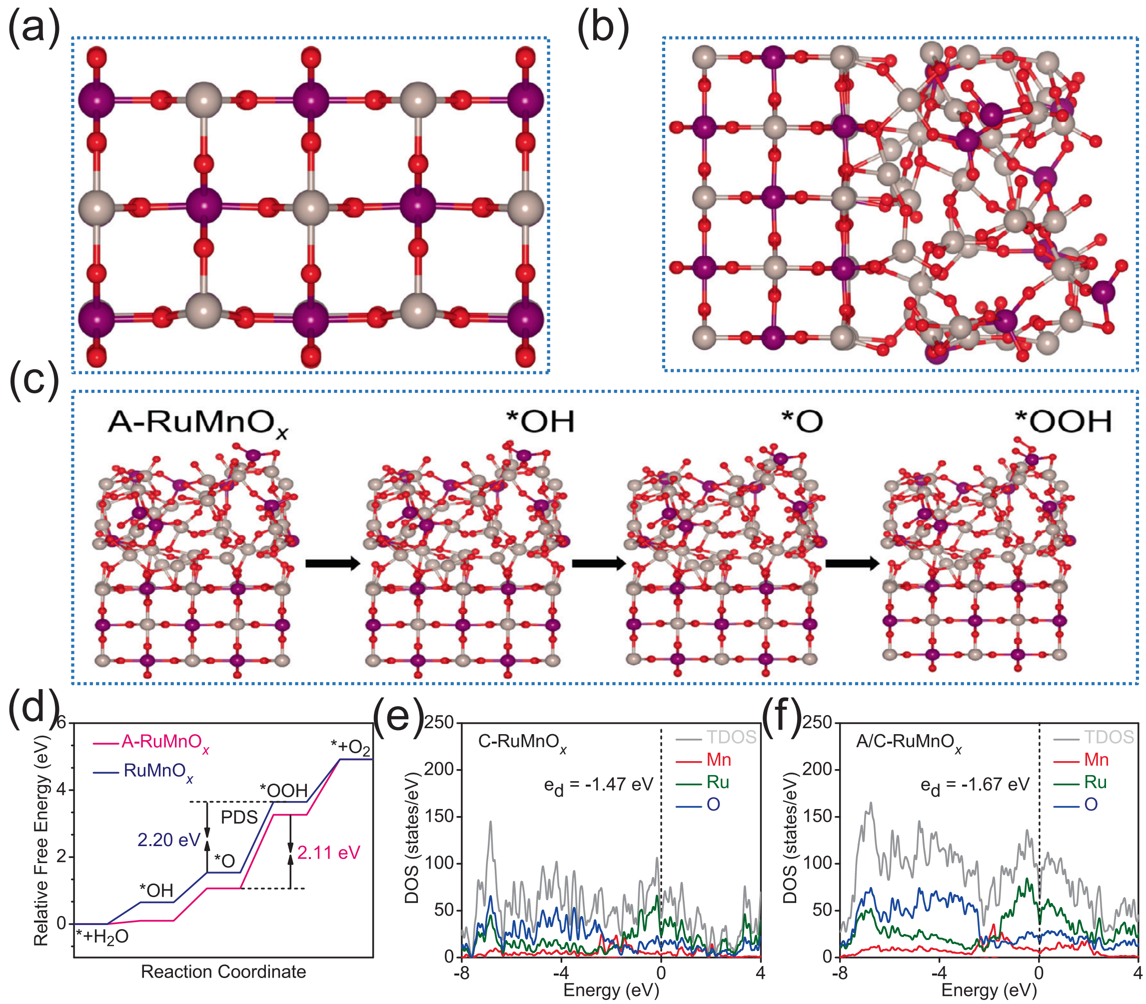

Fig. 5. (a, b) The model, (d) calculated free energy diagrams, (c) schematic representation of the OER process, and (e, f) PDOS diagrams of RuMnO$_x$ and A-RuMnO$_x$.

projected density of states (PDOS) analysis of A-RuMnO$_x$ and RuMnO$_x$, the *d*-band center of Ru in A-RuMnO$_x$ shows a negative shift away from the Fermi level, which confirms the weakening of the interaction with the oxygen-containing intermediates, as well as the positive role of the amorphous structure in lowering the OER energy barrier [8,59] (Fig. 5e and f).

4. Conclusions

In summary, we reported an amorphous-rich RuMnO$_x$ aerogel electrocatalyst with efficient and stable acidic OER capability. The amorphous structure not only optimized the electronic structure of Ru but also weakened the covalency of Ru–O in RuMnO$_x$, which prevented the lattice oxygen from participating in the OER, thus improving the OER activity and structural stability. As a result, the A-RuMnO$_x$ aerogel exhibited a very low overpotential of 145 mV at 10 mA cm^{-2}, high mass activity of 153 mA mg$_{Ru}^{-1}$ at 1.5 V vs. RHE, as well as long durability. In-situ techniques and DFT calculations revealed that RuMnO$_x$ followed the AEM pathway and optimized the energy barriers of the adsorbed intermediates, resulting in high OER stability and activity. This study provides new ideas for the development of Ir-free catalysts for the acidic OER, which will drive the development of electrocatalytic hydrogen production technology.

CRediT authorship contribution statement

Tao Zhao: Writing – original draft, Visualization, Methodology, Investigation. **Yunzhen Jia:** Visualization. **Qiang Fang:** Conceptualization. **Runxin Du:** Visualization. **Genyan Hao:** Writing – review & editing. **Wenqing Sun:** Writing – review & editing. **Guang Liu:** Formal analysis. **Dazhong Zhong:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization. **Jinping Li:** Writing – review & editing, Supervision, Formal analysis. **Qiang Zhao:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jechem.2024.12.053.

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